

Memo 12: Spatial Distribution Analysis for FVCOM-MP Cases D and E

FSK Settling and Effective Grain-Size Coupling, Days 10 and 30

FVCOM-MP WaterPACT Investigation

May 2026

Purpose

This memo records a new spatial-distribution comparison using the day-10 and day-30 `OUTPUT_d` and `OUTPUT_e` MAT files. The paired cases use the same cohesive FSK sediment setup and the same small MP size class. The primary question is whether turning on the MP-side equivalent floc or grain size treatment changes the MP fields.

Files and Command

The analysis inputs were:

- `FVCOM_MP_test_run/OUTPUT_d/spatial_distribution_v2_d_day10.mat`
- `FVCOM_MP_test_run/OUTPUT_e/spatial_distribution_v2_e_day10.mat`
- `FVCOM_MP_test_run/OUTPUT_d/spatial_distribution_v2_d_day30.mat`
- `FVCOM_MP_test_run/OUTPUT_e/spatial_distribution_v2_e_day30.mat`

The analysis was run with:

```
python FVCOM_MP_test_run/PYTHON/analyze_spatial_distribution_v2.py ^  
--days 10 30 --cases d e --diff-hi d --diff-lo e ^  
--output-base FVCOM_MP_test_run/PYTHON/output/spatial_distribution_v2_de
```

The script was updated so the difference pair is configurable. For this memo, difference maps and `diff_summary_stats.csv` use case `d` - case `e`. The outputs were written to:

- `FVCOM_MP_test_run/PYTHON/output/spatial_distribution_v2_de/day10`
- `FVCOM_MP_test_run/PYTHON/output/spatial_distribution_v2_de/day30`

The run generated 108 PNG maps across the two day folders.

Extraction Window

All four MAT files contain 24 output records. The day-10 files span simulation day 9.0000 to 9.9570, and the day-30 files span simulation day 29.0000 to 29.9570. These are final-24-hour averages before the target day endpoints, matching the v2 workflow used in Memo 09.

Paired Setup

Cases **d** and **e** are a cleaner pair than the Memo 09 **c**–**d** comparison because the active sediment setup is the same. Both cases use cohesive sediment, `SED_WSET_METHOD=FSK_CONC`, `SED_SD50=0.010` mm, and `SED_WSET=0.086` mm s^{−1}. The plastic setup is also the same except for the effective-floc switch:

Table 1: Active setup difference between cases **d** and **e**.

Item	Case d	Case e
Sediment file	<code>generic_sediment_d.inp</code>	<code>generic_sediment_e.inp</code>
Sediment type	cohesive	cohesive
Settling method	<code>FSK_CONC</code>	<code>FSK_CONC</code>
<code>SED_SD50</code>	0.010 mm	0.010 mm
<code>SED_WSET</code>	0.086 mm s ^{−1}	0.086 mm s ^{−1}
Plastic file	<code>generic_plastic_d.inp</code>	<code>generic_plastic_e.inp</code>
<code>PLAST_FLOC_EFFECTIVE</code>	T	F
MP axes	0.010, 0.002, 0.002 mm	0.010, 0.002, 0.002 mm

In the aggregation closure, the MP incorporation threshold uses the minimum MP dimension. For this setup, the relevant MP size is therefore 0.002 mm, or 2 μ m. The volume-equivalent MP diameter is approximately 3.42 μ m.

Key Spatial Statistics

Table 2 summarizes depth-averaged domain means for the fields most relevant to the FSK/effective-size question. Case **e** is the matched control with `PLAST_FLOC_EFFECTIVE=F`; case **d** has the effective-size coupling turned on.

Table 2: Depth-averaged domain means from the generated summary CSV files.

Day	Field	Case e	Case d	D/E or change
10	Sediment	9.555×10^{-1}	9.555×10^{-1}	no change
10	W_{floc} (m s ^{−1})	5.014×10^{-5}	1.533×10^{-4}	3.06×
10	$d_{\text{floc,eff}}$ (μ m)	16.01	26.41	1.65×
10	λ_a (s ^{−1})	5.887×10^{-1}	3.181×10^{-1}	0.54×
10	λ_d (s ^{−1})	1.394×10^{-1}	3.559×10^{-2}	0.26×
10	MP total (kg m ^{−3})	5.499×10^{-7}	6.037×10^{-7}	1.10×
10	MP aggregated (kg m ^{−3})	2.721×10^{-7}	5.307×10^{-7}	1.95×
10	MP dispersed (kg m ^{−3})	2.778×10^{-7}	7.302×10^{-8}	0.26×
10	Aggregated MP fraction	0.494	0.855	+0.361
10	Bottom MP mass (kg m ^{−2})	4.965×10^{-11}	7.886×10^{-10}	15.9×
30	Sediment	9.703×10^{-1}	9.703×10^{-1}	no change
30	W_{floc} (m s ^{−1})	8.105×10^{-5}	2.078×10^{-4}	2.56×
30	$d_{\text{floc,eff}}$ (μ m)	26.12	39.05	1.50×
30	λ_a (s ^{−1})	5.057×10^{-1}	2.705×10^{-1}	0.53×
30	λ_d (s ^{−1})	1.199×10^{-1}	2.746×10^{-2}	0.23×
30	MP total (kg m ^{−3})	1.041×10^{-6}	1.315×10^{-6}	1.26×

Day	Field	Case e	Case d	D/E or change
30	MP aggregated (kg m ⁻³)	5.143×10 ⁻⁷	1.140×10 ⁻⁶	2.22×
30	MP dispersed (kg m ⁻³)	5.264×10 ⁻⁷	1.758×10 ⁻⁷	0.33×
30	Aggregated MP fraction	0.494	0.863	+0.369
30	Bottom MP mass (kg m ⁻²)	9.516×10 ⁻¹¹	1.369×10 ⁻⁹	14.4×

Main Findings

The sediment fields are unchanged

The **d-e** difference statistics show exactly zero difference in `sed_davg`, `sed_surf`, and `sed_bot` for both days. This is important: the analysis isolates the MP-side effective-size switch rather than a sediment-regime change. It also means that the MP response below is not caused by a different sediment concentration field.

The FSK equivalent size is active and visible

With `PLAST_FLOC_EFFECTIVE=T`, case **d** uses the active FSK settling signal to compute the floc size used by the MP interaction path. The depth-averaged diagnostic settling speed is about 3.06 times case **e** on day 10 and 2.56 times case **e** on day 30. The corresponding domain-mean diagnostic Stokes-equivalent floc size increases from 16.0 to 26.4 μm on day 10 and from 26.1 to 39.1 μm on day 30.

Case **e** still has a diagnostic `d_floc_eff` field in the MAT files because the extractor back-calculates it from the saved settling velocity. However, with `PLAST_FLOC_EFFECTIVE=F`, the model's aggregation kernel uses the static sediment size `SED_SD50=0.010` mm inside the MP-floc interaction.

The small MP class remains below the incorporation threshold

The aggregation code uses the minimum MP dimension for the Wu-style incorporation threshold, so the relevant MP size is 2 μm . For a 10 μm static sediment size, the threshold

$$d_{\text{MP,max}} = -0.0002d_{\text{floc}}^2 + 0.36d_{\text{floc}}$$

is about 3.58 μm . For the case-d mean effective floc sizes of 26–39 μm , the threshold is about 9–14 μm . The 2 μm MP class is therefore allowed to aggregate in both cases, but the effective-size case has a much larger margin. This is different from Memo 09, where the 100 μm MP class was too large for the small effective flocs.

Aggregation rate decreases, but aggregated MP fraction increases

Turning on the effective-size treatment reduces the domain-mean aggregation rate by about 46% on both days. It also reduces the disaggregation rate more strongly: by about 74% on day 10 and 77% on day 30. The lower disaggregation rate dominates the state variables: the aggregated MP fraction rises from about 0.494 in case **e** to 0.855–0.863 in case **d**. The aggregated MP concentration approximately doubles, while the dispersed MP concentration drops to roughly one-quarter to one-third of the control.

Bottom MP mass is highly sensitive to the switch

The bottom MP mass is much larger when the effective-size treatment is on: about 15.9 times the control on day 10 and 14.4 times the control on day 30. The bottom aggregated MP mass remains zero in the generated summaries, so this increase appears in the bottom dispersed MP mass diagnostic. The result still indicates a strong transport and deposition response associated with the effective-size treatment.

Interpretation

The answer to the main question is yes: turning on the equivalent grain or floc size treatment has a clear impact in this paired test. Because cases **d** and **e** have identical sediment concentration fields, the impact is attributable to the MP-side use of the FSK-derived settling and equivalent-size information.

The mechanism is not a simple increase in every aggregation metric. The effective-size treatment makes the MP interaction use larger effective flocs and a stronger settling signal. For the very small MP class used here, the MP remains comfortably below the floc incorporation threshold, so aggregation is not shut off. Instead, the rate diagnostics show lower instantaneous aggregation but much lower disaggregation, which shifts the modeled MP state toward aggregated suspended mass and much larger bottom mass.

This paired **d-e** comparison is therefore a useful production test for the new FSK/effective-size pathway. It shows that the new pathway is active, spatially resolved, and dynamically important for small MPs.

Recommended Next Checks

1. Inspect the day-30 difference maps for `lambda_a_davg`, `lambda_d_davg`, `agg_fraction_davg`, `mp1_agg_davg`, and `bot_mass`; these are the fields where the switch has the clearest spatial footprint.
2. Add a direct model output diagnostic for the floc size actually used by the aggregation kernel and for $d_{\text{MP,max}}$. The current MAT files include a useful post-processed equivalent size, but the aggregation-kernel size is still inferred from the switch logic.
3. Repeat the same paired analysis for day 90 once `spatial_distribution.v2_d/e_day90.mat` files are available.